Near-real-time monitoring of global ocean carbon sink

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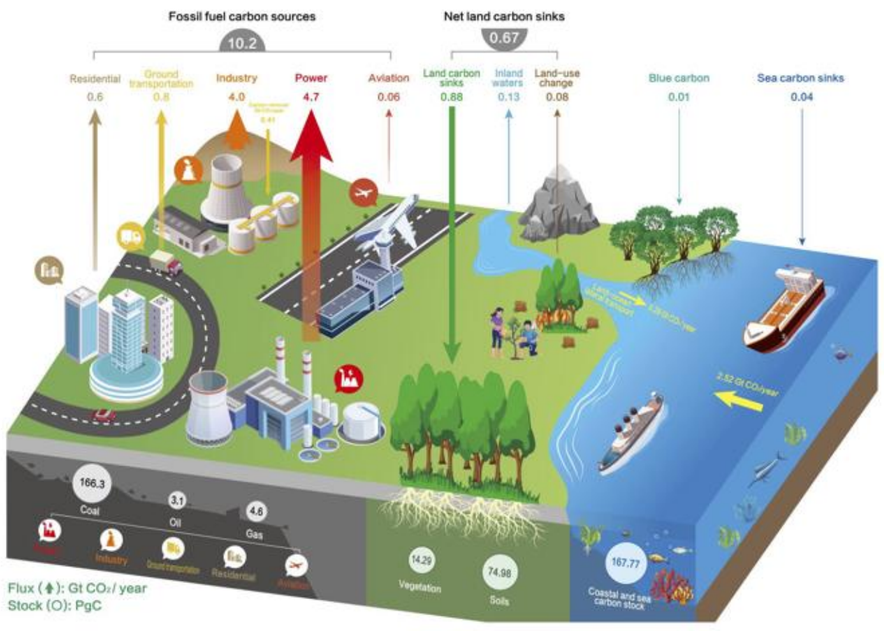
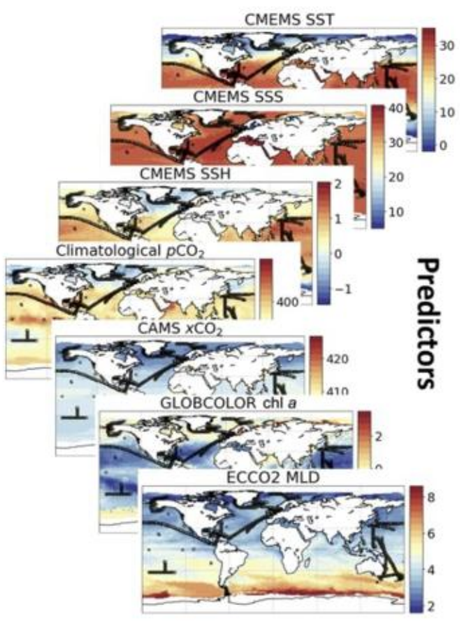
Introduction



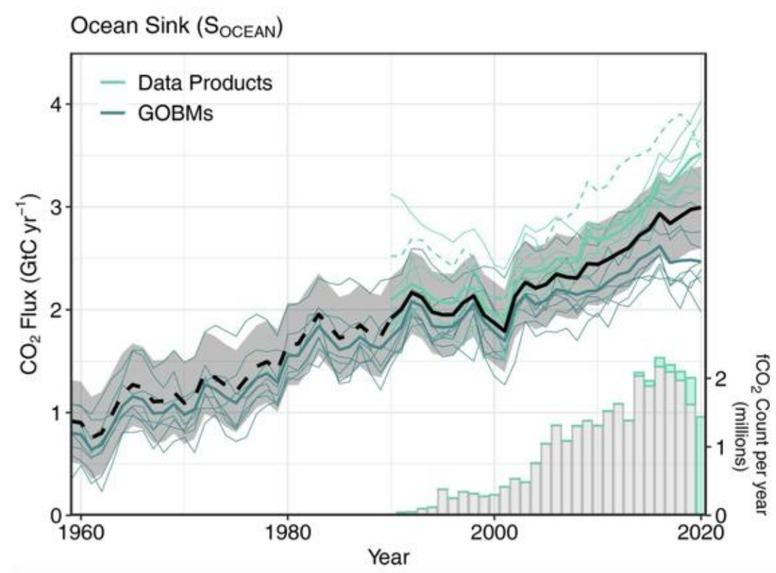
We present Carbon Monitor Ocean (CMO-NRT), a novel dataset providing near-real-time monthly gridded estimates of global surface ocean fugacity of $CO_2(fco_2)$and ocean-atmosphere CO_2 flux from January 2022 to July 2023. Leveraging Convolutional Neural Networks and semi-supervised learning, our highly accurate models enhance timely climate change mitigation efforts.



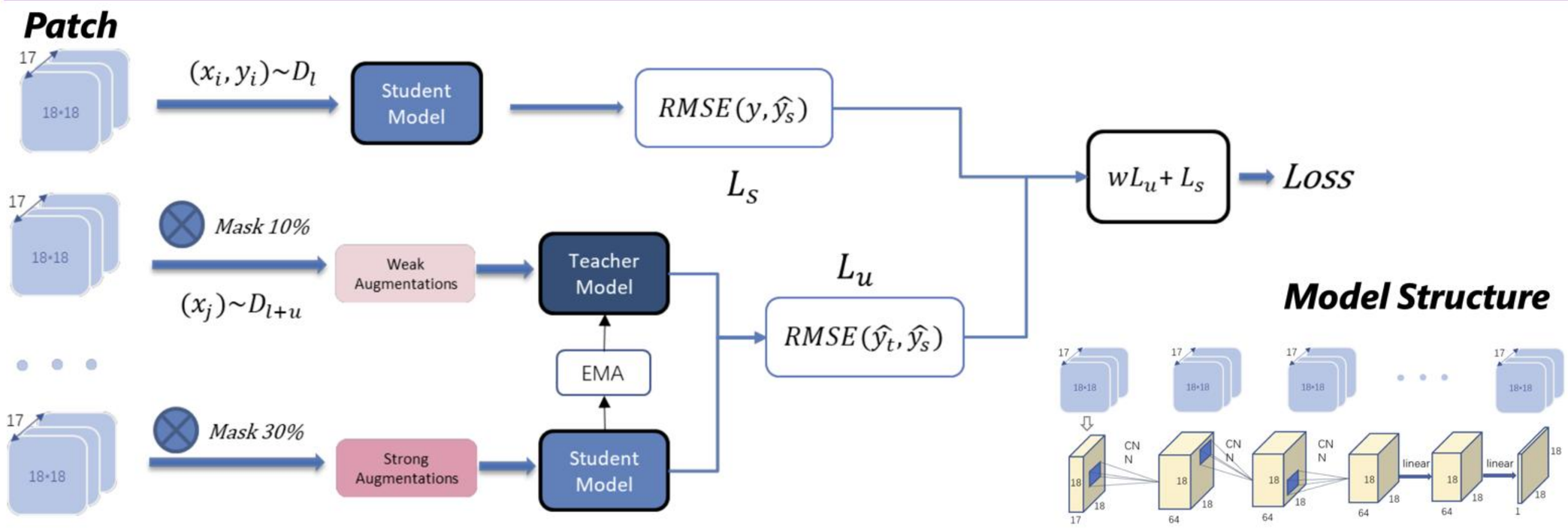
Environmental factors
Predictors

Global ocean carbon sink
Target

- **Environmental factors influencing and reflecting oceanic carbon sink variations**
- **Reconstructing global oceanic carbon sink using ML methods with global environmental data**

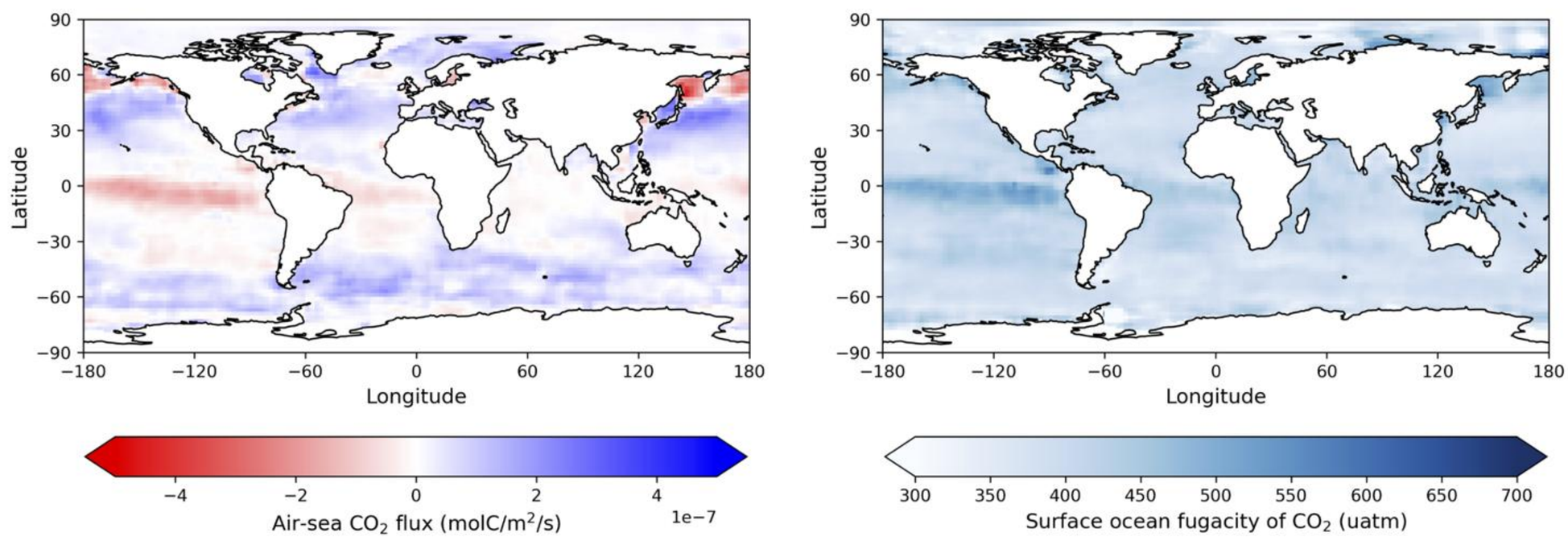


Model



Result

Our models maintain accuracy with errors below a **5%** threshold(NRMSE), most models have losses below **2%**



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